



5.3.4 Solving $Ax = y$ via LU factorization with pivoting



05.3.4 Solving with LU factorization with pivoting

fit width



Given nonsingular matrix $A \in \mathbb{C}^{m \times n}$, the above discussions have yielded an algorithm for computing permutation matrix P , unit lower triangular matrix L and upper triangular matrix U such that $PA = LU$. We now discuss how these can be used to solve the system of linear equations $Ax = y$.

Starting with

$$Ax = b$$

where nonsingular matrix A is $n \times n$ (and hence square),

- Overwrite A with its LU factorization, accumulating the pivot information in vector p :

$$[A, p] := \text{LU piv}(A).$$

A now contains L and U and $\tilde{P}(p)A = LU$.

- We notice that $Ax = b$ is equivalent to $\tilde{P}(p)Ax = \tilde{P}(p)b$. Thus, we compute $y := \tilde{P}(p)b$. Usually, y overwrites b .
- Next, we recognize that $\tilde{P}(p)Ax = y$ is equivalent to $L(\underbrace{Ux}_z) = y$. Hence, we

can compute z by solving the unit lower triangular system

$$Lz = y$$

and next compute x by solving the upper triangular system

$$Ux = z.$$